

Grid Equipment Failure Advisor

From hidden equipment risk to prioritized grid action

Combine asset health, weather, and historical incidents to help grid operators act before failures become outages.

IBM Bob AI Hackathon | Team Fantastic4

Why grid failures hurt

Who is affected

- Utilities & grid operators
- Maintenance teams & field crews
- Millions of electricity customers

What happens

Transformers and substations can fail unexpectedly, causing blackouts and complex emergency response.

Operational gap

Maintenance is often calendar-based while actionable signals exist across sensors, weather, and incident data. Costly outages can follow — e.g., \$1M+/hour contextual risk.

The data exists. The decision is fragmented.



Asset health

Temperature, vibration, oil quality, partial discharge — sensor abnormalities.



Weather

Storms and forecast conditions that increase failure risk.



Historical incidents

Past failures, patterns, and asset history by district.

No unified risk picture → no impact-aware prioritization → reactive crew deployment

❏ Operators need one view: What is at risk? Where will failure hurt most? What to do first?

A decision-support layer for grid operations

1. Ingest

Asset sensors, weather feeds,
incident records

2. Assess

Compute asset & district risk from
combined signals

3. Prioritize

Rank by failure risk + grid impact
(customers, criticality)

4. Act

Bob-assisted maintenance plans + crew dispatch guidance

Not just predicting risk — turning risk into an operational response.

One dashboard. Four operational questions.



Where is risk concentrated?

District risk overview & Gujarat map



Which asset needs attention?

Asset risk ranking and detail view



Why is it risky?

Contributing factors + Bob-generated explanation



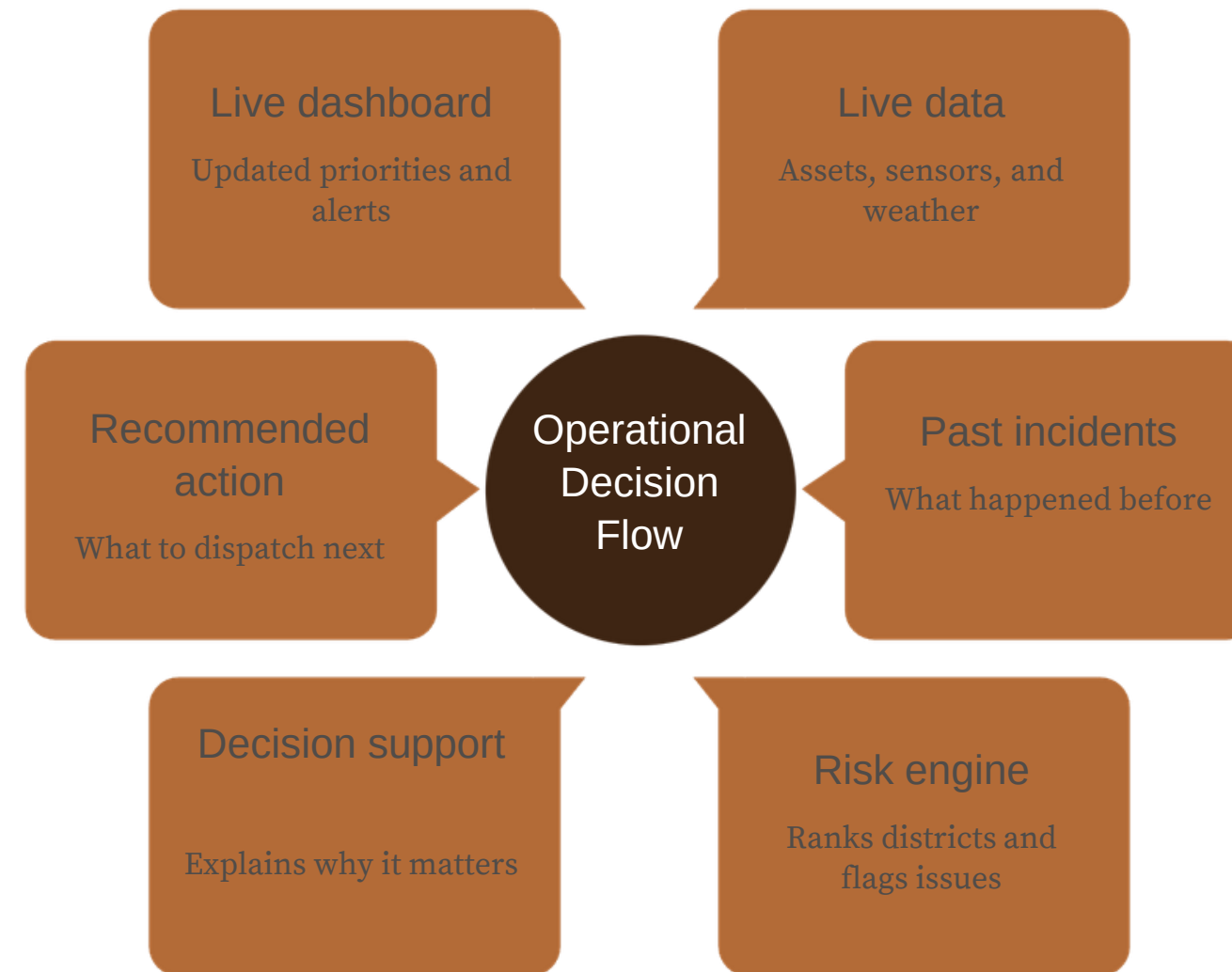
What should the team do?

Maintenance priorities, dispatch plan, crew recommendations



Slides use the actual screenshots from the repository frontend — authentic product evidence.

From data to dispatch.



Use the dashboard to see what needs attention, why it is a priority, and what action to take next. Updates arrive live so crews can respond quickly and stay focused on the highest-risk issues.

Where IBM Bob adds intelligence



Risk explanation

Bob turns scored data + factors into plain-language explanations for operators.



Operational follow-up

Interactive Q&A about assets, context-aware answers.



Plan generation

Structured maintenance & dispatch plan with priority, crew type, location, ETA, pre-positioning advice.



Controlled reliability

Application validates Bob output; deterministic rule-based fallback available when Bob is unavailable or invalid.

A critical asset becomes an actionable priority

- 1.**Detect** — Asset shows elevated risk on the Gujarat map (live sensor feed).
- 2.**Explain** — Operator opens asset: sensors, weather, incident history detailed.
- 3.**Prioritize** — System ranks asset by risk, estimated customers affected, and criticality.
- 4.**Respond** — Bob generates maintenance & crew dispatch recommendation (urgency, action, location, pre-positioning).

 Demo steps use real dashboard views and implemented Bob integration from the repository.

From reactive maintenance to proactive grid resilience

What the prototype demonstrates

- Unified asset + district risk view for Gujarat
- Impact-aware prioritization (risk score + customers affected + criticality)
- Explainable risk insights via IBM Bob
- Actionable maintenance plans and crew pre-positioning
- Live sensor/status streaming in the frontend

What this could become

- SCADA / IoT / asset-management integration
- More historical failure data and validated predictive models
- Real-time severe-weather alerts and tighter operational automation
- Crew availability, travel time, spare-parts constraints, and feedback loops
- Utility-wide deployment with audit trails and human approval gates

Prevent outages. Restore faster.

Resilience improves when operators know what may fail, what matters most, and what to do next.



See risks early

Sensor, weather, and history in one view



Prioritize impact

Rank by risk and customers affected



Respond faster

Bob-guided plans and crew direction

Grid Equipment Failure Advisor — An AI layer for proactive maintenance and rapid response

See it in action — demo available on request